# Chapter summary

Familiarising yourself with **items in green** is important for effective revision.
Paying attention to items in the 📝 boxes is helpful for understanding the concept(s).

## 11.1 Price Ceiling (Maximum Price Control)

### Meaning of a price ceiling

- A **price ceiling** (maximum price control) is the upper limit imposed by the government on the price sellers can charge for a good or a service.
- A price ceiling is **effective only if it is set below the equilibrium price**.

### Effects of an effective price ceiling

- When an effective price ceiling is imposed,
  - the price decreases;
  - the quantity transacted decreases;
  - a shortage/an excess demand exists;
  - total revenue decreases.
- An illegal black market may exist in which the price is usually higher than the price ceiling.
- As a shortage exists under an effective price ceiling, non-price competition is used to allocate goods:
  - on a first-come-first-served basis
  - by consumers' abilities
  - by consumers' needs
  - by preference of the seller
  - by drawing lots

## 11.2 Price Floor (Minimum Price Control)

### Meaning of a price floor

- A **price floor** (minimum price control) is the lower limit imposed by the government on the price sellers can charge for a good or a service.
- A price floor is **effective only if it is set above the equilibrium price**.

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## Effects of an effective price floor

- When an effective price floor is imposed,
  - the price increases;
  - the quantity transacted decreases;
  - a surplus/an excess supply exists;
  - total revenue may increase, decrease or remain unchanged, **depending on the price elasticity of demand for the good**:

| Price elasticity of demand | Total revenue |
|---|---|
| elastic ($1 < E_d < \infty$) | decreases |
| inelastic ($0 < E_d < 1$) | increases |
| unitarily elastic ($E_d = 1$) | remains unchanged |

- A black market may exist in which the price is usually lower than the price floor.
- As an excess supply exists under an effective price floor, sellers may offer buyers illegal cash rebates or non-monetary benefits.



The black market price can be as low as the price that corresponds to the quantity transacted on the supply curve.

# 11.3 Quota

## Meaning of a quota

- A **quota** is a limit on the maximum quantity of goods that sellers can sell in the market.
- A quota is **effective only if it is set below the equilibrium quantity**.



An increase (A decrease) in quota leads to an increase (a decrease) in the supply.

## Effects of an effective quota

- When an effective quota is imposed,
  - the supply curve below the quota limit is upward-sloping and will be 'kinked' and becomes vertical at the quota limit;
  - the price increases;
  - the quantity transacted decreases;
  - total revenue may increase, decrease or remain unchanged, **depending on the price elasticity of demand for the good**:

| Price elasticity of demand | Total revenue |
|---|---|
| elastic ($1 < E_d < \infty$) | decreases |
| inelastic ($0 < E_d < 1$) | increases |
| unitarily elastic ($E_d = 1$) | remains unchanged |

  - quality of the goods may improve.

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# 11.4 Tax on Goods and Services (Sales Tax)

## Meaning of a sales tax

- A **sales tax** is a tax levied on the sale or purchase of goods and services.
- There are two ways of levying a sales tax:
  - **Per-unit tax**: a tax in which a fixed amount is levied on every unit of the good sold
  - ***Ad valorem* tax**: a tax that is usually levied as a percentage of the price of a good

## Effects of a per-unit sales tax

- When a per-unit sales tax is levied,
  - as the cost of production increases, supply decreases and the supply curve shifts vertically upwards;
  - the price increases;
  - the quantity transacted decreases;
  - total revenue (including tax) may increase, decrease or remain unchanged, **depending on the price elasticity of demand for the good**:

| Price elasticity of demand | Total revenue (including tax) |
|---|---|
| elastic ($1 < E_d < \infty$) | decreases |
| inelastic ($0 < E_d < 1$) | increases |
| unitarily elastic ($E_d = 1$) | remains unchanged |

  - total revenue (net of tax) decreases;
  - the government receives tax revenue.

For a perfectly inelastic demand or a perfectly elastic supply, the increase in price due to a per-unit tax equals the amount of tax. In other cases, the increase in price is less than the amount of tax.

## Share of the tax burden

$$\text{Total tax burden} = \text{Per-unit tax} \times \text{Quantity transacted after tax}$$

$$\text{Consumers' tax burden} = (\text{Price after tax} - \text{Price before tax}) \times \text{Quantity transacted after tax}$$

$$\text{Producers' tax burden} = [\text{Price before tax} - (\text{Price after tax} - \text{Per-unit tax})] \times \text{Quantity transacted after tax}$$

- The more elastic party will have a smaller share of the tax burden.

| Situation | Result |
|---|---|
| $E_d > E_S$ | Consumers' tax burden < Producers' tax burden |
| $E_d < E_S$ | Consumers' tax burden > Producers' tax burden |
| $E_d = E_S$ | Consumers' tax burden = Producers' tax burden |

Producers bear the entire tax burden when the demand is perfectly elastic or the supply is perfectly inelastic.

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# 11.5 Subsidy on Goods and Services

## Meaning of a subsidy

- A **subsidy** on goods and services is a financial aid provided by the government on the sale or purchase of goods and services.

## Effects of a per-unit subsidy

- When a per-unit subsidy is provided,
  - as the cost of production decreases, supply increases and the supply curve shifts vertically downwards;
  - the price decreases;
  - the quantity transacted increases;
  - total revenue (excluding subsidy) may increase, decrease or remain unchanged, **depending on the price elasticity of demand for the good**:

| Price elasticity of demand | Total revenue (excluding subsidy) |
|---|---|
| elastic ($1 < E_d < \infty$) | increases |
| inelastic ($0 < E_d < 1$) | decreases |
| unitarily elastic ($E_d = 1$) | remains unchanged |

  - total revenue (including subsidy) increases.

## Share of the subsidy

$$\text{Total amount of subsidy} = \text{Per-unit subsidy} \times \text{Quantity transacted after the subsidy}$$

$$\text{Consumers' share of the subsidy} = \left(\text{Price before the subsidy} - \text{Price after the subsidy}\right) \times \text{Quantity transacted after the subsidy}$$

$$\text{Producers' share of the subsidy} = \left[\left(\text{Price after the subsidy} + \text{Per-unit subsidy}\right) - \text{Price before the subsidy}\right] \times \text{Quantity transacted after the subsidy}$$

- The more elastic party will have a smaller share of the subsidy.

| Situation | Result |
|---|---|
| $E_d > E_S$ | Consumers' share of the subsidy < Producers' share of the subsidy |
| $E_d < E_S$ | Consumers' share of the subsidy > Producers' share of the subsidy |
| $E_d = E_S$ | Consumers' share of the subsidy = Producers' share of the subsidy |

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